

AGUSTÍN HERRERA

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PROFESSIONAL SUMMARY

Computer Science Engineer (Pontificia Universidad Católica de Chile) with 2+ years of professional experience designing and operating scalable distributed systems on AWS. Proven track record building ML inference pipelines, multi-step orchestration workflows (Step Functions, SQS, ECS), multi-agent AI applications, and full-stack systems with modern TypeScript/JavaScript frameworks. Passionate about generative AI and eager to leverage LLMs to power dynamic, personalized creative experiences at Amazon scale.

PROFESSIONAL EXPERIENCE

DevOps / Software Development Engineer

Dec 2024 – Present

diio *Santiago, Chile*

- Architected and deployed an end-to-end **ML inference pipeline for audio transcription and speaker diarization** orchestrating **AWS Step Functions, Lambda, SQS, ECS, and S3**; replaced a third-party vendor and **cut costs by 60%** while improving throughput and reliability.
- Built **Spark**, a multi-agent AI sales assistant powered by **Google Vertex AI Reasoning Engine** (agent loop pattern: plan → tool call → result → continue reasoning). Implemented the **MCP server in Ruby on Rails** exposing 6 tools — RAG over Elasticsearch, text-to-DSL query generation, CRM integrations, and chart generation — while the LLM reasoning runs externally on Google Cloud. Demonstrated clean separation of concerns: Google owns “**when to call**”, Rails owns “**how to execute**”.
- Engineered the **RAG pipeline** for Spark: embeds queries via **OpenAI text-embedding-3-large**, runs hybrid **BM25 + KNN vector search** on Elasticsearch, and returns ranked conversation snippets (up to 35 conversations, capped at 100K tokens) with rich XML context blocks for the LLM to synthesize answers.
- Implemented **dynamic auto-scaling of ECS Sidekiq workers** using **CloudWatch alarms** triggered by real-time queue depth metrics, reducing compute costs without sacrificing throughput.
- Designed and deployed an **observability stack** with **CloudWatch, Prometheus, and Grafana** (hosted on EC2), surfacing key business and infrastructure metrics and enabling proactive incident detection across the production environment.
- Migrated production database from **RDS PostgreSQL to distributed Citus**, enabling horizontal scaling and improving analytical query performance by **10%**.
- Optimized **CI/CD pipelines with GitHub Actions**, reducing deployment time by **50%** through parallelization, build caching, and matrix builds.
- Built an AI-powered nightly **security scanning pipeline** on on-premise **NVIDIA Jetson hardware** that automatically opens **Pull Requests with LLM-generated vulnerability fixes and justifications**.

Software Development

Aug 2023 – Dec 2023

Lolocar.com *Santiago, Chile*

- Developed scalable **RESTful APIs** for tenant toll-charge calculation, integrated with existing backend systems using **Ruby on Rails**.
- Designed and implemented internal **admin dashboards** for real-time data management, improving operational efficiency for the ops team.
- Led application deployment to **Google Cloud Platform**, improving deployment reliability and speed by **20%**.

EDUCATION

Pontificia Universidad Católica de Chile

Mar 2020 – Dec 2024

B.S. in Computer Science Engineering

Santiago, Chile

- Major:** Software Engineering | **Minor:** Scientific & Technological Foundations of Computing

TECHNICAL SKILLS

Languages:

Python, TypeScript / JavaScript, Go, Ruby, Java, C / C++, C#, Bash, R

Cloud & Infrastructure:

AWS (EC2, ECS, ECR, S3, IAM, Lambda, Step Functions, SQS, CloudWatch, RDS), Docker, GCP

AI / ML & Agents:

Multi-agent systems (Vertex AI Reasoning Engine), LLM automation, RAG pipelines,

OpenAI embeddings, hybrid vector search (BM25 + KNN), GenAI integration

Observability:

CloudWatch, Prometheus, Grafana (EC2-hosted), ECS auto-scaling with CloudWatch alarms

Backend & Frameworks:

Ruby on Rails, Nest.js, Next.js, React, Vue, Svelte, Angular

Databases & Search:

PostgreSQL, Citus (distributed), Redis, Elasticsearch, SQL

CI/CD & DevOps:

GitHub Actions, Bash scripting, Linux, build caching, matrix builds

Languages (Human):

Spanish (Native), English (C2 – Proficient)